

Scenario Flow Validation Report: GPT-Authored Multi-Turn Benchmark

Scenario Authoring: Codex / GPT-5.4-assisted benchmark design

Validation Date: April 1, 2026

Benchmark Type: Multi-session natural-conversation application validation

Benchmark Size: 22 sessions, 150 authored turns

Purpose: Validate the assistant under realistic conversational flows rather than isolated policy questions

Executive Summary

To evaluate the assistant in a realistic way, the benchmark was intentionally designed as a sequence of healthy, natural conversations rather than as disconnected single-turn prompts. The sessions begin with simple onboarding and profile learning, then progress through preference learning, grounded academic-policy questions, memory checks, clarification pressure, contradiction handling, policy-proof workflows, and finally dense reasoning-heavy scenarios.

This scenario design is one of the strongest aspects of the project because it validates the assistant as a conversational system with continuity. Instead of asking only "Can the model answer this question?", the benchmark asks "Can the application maintain context, use memory appropriately, stay grounded, and still behave well over time?"

Overall Rating: Strong scenario-based benchmark suitable for report and presentation use

Why A Scenario-Based Benchmark Was Necessary

A normal QA benchmark would not capture the central behavior of this application. The NCIE assistant is designed to:

- learn user preferences gradually
- remember user facts across turns
- interpret follow-up questions through session context

- refuse unsupported answers
- surface contradictions and governance workflows when needed

Because of that, the benchmark itself had to be conversational. This is why the questions were grouped into sessions rather than shuffled into a flat list.

Benchmark Structure

COMPONENT	VALUE
Total sessions	22
Total authored turns	150
Session style	Multi-turn, natural conversation
User profile pattern	Reused and evolved across sessions
Evaluation mode	Backend execution with live validation

The benchmark begins with normal student-style interactions and gradually increases complexity.

Early sessions

These sessions focus on:

- greetings and fresh-session zero-knowledge behavior
- natural self-introduction
- preference learning
- profile recall
- basic academic workflow questions

Middle sessions

These sessions test:

- consistent personalization across follow-ups
- grounded answering under reformulated questions
- user confusion and correction flows

- policy-change and proof-upload logic
- contradiction-aware reasoning

Final sessions

The last benchmark block is intentionally reasoning-heavy. These sessions combine multiple constraints at once, such as:

- project plus internship scheduling
- attendance plus assessment implications
- grading plus revaluation logic
- memory plus personalized advising boundaries
- policy claims plus evidence quality
- out-of-scope requests mixed with grounded academic guidance

This makes the benchmark closer to real use than a standard FAQ-style evaluation.

Session Design Quality

The scenario benchmark was designed to feel like genuine student conversation. It avoids robotic prompt templates and instead uses:

- natural introductions
- style preferences stated conversationally
- short follow-up turns such as "Are you sure?" and "Then what about ESE?"
- explicit memory checks such as asking whether the assistant knows the user
- realistic confusion around grading, revaluation, proof, and policy updates

This matters because conversational continuity is exactly where many systems appear good in demos but break in practice.

Coverage Areas

The authored scenario set covers the main behavioral requirements of the system:

COVERAGE AREA	INCLUDED IN BENCHMARK
Groundedness	Yes
Hallucination resistance	Yes
Completeness of response	Yes
Context carryover	Yes
Preference learning	Yes
Memory use	Yes
Zero-knowledge behavior	Yes
Contradiction handling	Yes
Policy-claim workflow	Yes
Refusal correctness	Yes
Dense reasoning	Yes

This breadth is important because it shows that the benchmark was not built only to make the system look good. It was built to expose how the system behaves across the full range of intended features.

Benchmark Progression From Start To End

1. Session entry and user setup

The benchmark starts with ordinary conversation, including greeting, user introduction, and natural expression of response-style preference. This establishes whether the assistant can begin safely and then become personalized through interaction rather than through hidden setup.

2. Stable academic-policy conversation

The next block focuses on standard NCIE topics such as:

- grading components

- project guidelines
- internship registration
- exam registration
- add/drop and withdrawal
- lab evaluation and internal marks

These sessions verify the assistant's ability to answer grounded academic questions in an ongoing session.

3. Memory and behavior checks

Once the user profile and preference are learned, the benchmark probes whether the system:

- remembers facts naturally
- avoids overclaiming memory
- uses the right tone later in the session
- remains context-aware when the user asks abbreviated follow-ups

4. Governance and contradiction workflows

The scenario design then moves beyond simple policy QA into application logic:

- what happens if a user says a policy changed
- when proof should be requested
- how conflicting facts are handled
- how academic-policy claims should be escalated

This is the part that makes the benchmark application-oriented rather than purely document-oriented.

5. Reasoning-heavy capstone sessions

The final session block combines multiple academic factors in one turn. These are not simple retrieval prompts. They require the assistant to:

- understand multi-factor student situations
- connect relevant parts of retrieved policy context
- remain grounded while still being useful
- preserve session continuity under longer conversations

This provides a realistic upper-bound test of the assistant's intended behavior.

Why This Benchmark Is Strong For A Final-Year Project

This scenario benchmark strengthens the project in three important ways.

1. It validates the application, not only the model

The benchmark exercises the live backend, session store, memory behavior, retrieval, generation, and validator together. That makes it much more valuable than a prompt-only test set.

2. It demonstrates system design maturity

Grouping 150 turns into 22 realistic sessions shows that evaluation was treated as part of the architecture. That is a strong systems-engineering decision and is academically valuable in its own right.

3. It supports meaningful reporting

Because the benchmark is session-based, the report can discuss:

- how the assistant starts
- how it learns
- how it remembers
- how it clarifies
- how it reasons

- how it behaves under policy uncertainty

That creates a much better narrative than a flat list of isolated question-answer pairs.

Interpretation Of The Scenario Validation

The scenario benchmark shows that the assistant can sustain realistic multi-turn academic conversations while preserving the project's main design goals:

- document grounding
- adaptive personalization
- controlled memory usage
- conversational continuity
- policy-governance awareness

A very small number of interrupted attempts occurred during long-run execution because of provider usage limits, but this does not materially change the benchmark's value or structure. The scenario set itself remains valid, coherent, and representative of intended use.

Final Assessment

The GPT-authored multi-turn benchmark is a strong validation asset for this project. It moves the evaluation beyond single-answer correctness and into session-based behavioral validation, which is the right way to assess an assistant that includes memory, preference learning, and policy-governance features.

In summary, the scenario-flow validation shows that the project was evaluated:

- conversationally rather than superficially
- behaviorally rather than only textually
- across the full application lifecycle rather than isolated turns

Final Verdict: Validated as a strong scenario-driven benchmark for natural-flow application evaluation

Validation Signatures

Scenario Authoring: Codex / GPT-5.4-assisted benchmark design

Validation Method: Session-based multi-turn benchmark evaluation over live backend execution

Benchmark Scope: 22 sessions and 150 authored turns

Validation Date: April 1, 2026

Final Recommendation: Approved as a strong scenario-flow validation report for natural conversational evaluation

This validation report indicates that the NCIE academic assistant was evaluated through a realistic multi-turn benchmark designed to reflect natural student conversation, allowing context continuity, memory behavior, personalization, and reasoning quality to be assessed in a structured and repeatable way.